Enzyme Linked Immunosorbent Assay

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Introduction

Enzyme immunoassays (EIAs) use the catalytic properties of enzymes to detect and quantify immunologic reactions. Enzyme-linked immunosorbent assay (ELISA) is a heterogeneous EIA technique used in clinical analyses.[1] In this type of assay, one of the reaction components is nonspecifically adsorbed or covalently bound to the surface of a solid phase, such as a microtiter well, a magnetic particle, or a plastic bead. This attachment facilitates the separation of bound and free-labeled reactants.[2]

In the most common approach to using the ELISA technique, an aliquot of sample or calibrator containing the antigen (Ag) to be quantified is added to and allowed to bind with a solid-phase antibody (Ab). After washing, an enzyme-labeled antibody is added and forms a "sandwich complex" of solid-phase Ab-Ag-Ab enzyme. Unbound antibody is then washed away, and enzyme substrate is added. The amount of product generated is proportional to the quantity of antigen in the sample.[1]

Specific antibodies in a sample can also be quantified using an ELISA procedure in which antigen instead of antibody is bound to a solid phase. The second reagent is an enzyme-labeled antibody specific to the analyte antibody.[3] In addition, ELISA assays have been used extensively to detect antibodies to viruses and autoantigens in serum or whole blood. In addition, enzyme conjugates coupled with substrates that produce visible products have been used to develop ELISA-type assays with results that can be interpreted visually. Such assays are very useful in screening, point-of-care, and home testing applications.[4]

Etiology and Epidemiology

Two different research teams simultaneously invented the direct ELISA; scientists Engvall and Perlman, and scientists Van Weemen and Schuurs. The ELISA was developed by the modification of the radioimmunoassay (RIA). This was done by conjugating tagged antigens and antibodies with enzymes rather than radioactive iodine 125. The new method was first employed in determining the levels of IgG in rabbit serum. Within the same year, scientists quantified human chorionic gonadotropin in urine using horseradish peroxidase. Since then, the ELISA method has been used in many different applications and has become a routine laboratory research and diagnostic method worldwide.[1]

The first ELISA methodology involved chromogenic reporter molecules and substrates in generating observable color change that monitors the presence of antigen. Further advancement in the ELISA technique led to the development of fluorogenic, quantitative PCR, and

electrochemiluminescent reporters to generate signals.[5] However, some of these techniques do not rely on using enzyme-linked substrates but non-enzymatic reporters that utilize the principle of ELISA.[6]

The latest development, in 2012, was an ultrasensitive enzyme-based ELISA that manipulates nanoparticles as chromogenic reporters. This technique can generate a color signal visible to the naked eye, with a blue color indicating positive results and red color indicating negative results. However, this method is qualitative and can determine only the presence or absence of an analyte and not its concentration.[7]

Specimen Requirements and Procedure

ELISAs are performed in polystyrene plates, typically 96-well plates coated to bind protein strongly.[2] Depending on the ELISA type, testing requires a primary and/or secondary detection antibody, analyte/antigen, coating antibody/antigen, buffer, wash, and substrate/chromogen.[3] The primary detection antibody is a specific antibody that only binds to the protein of interest. In contrast, a secondary detection antibody is a second enzyme-conjugated antibody that binds to a primary antibody that is not enzyme-conjugated.[4]

There are four main general steps to completing an ELISA immunoassay. These steps are:

1. Coating (with either antigen or antibody)
2. Blocking (typically with the addition of bovine serum albumin [BSA])
3. Detection
4. Final read

Detection is carried out by adding a substrate that can generate a color. There are many substrates available for use in ELISA detection. However, the substrates most commonly used are horseradish peroxidase (HRP) and alkaline phosphatase (AP).[3] The substrate for HRP is hydrogen peroxide, resulting in a blue color change. AP measures the yellow color of nitrophenol after room temperature incubation periods of 15 to 30 minutes and usually uses p-Nitrophenyl-phosphate (pNPP) as its substrate.[2]

Between each of the above four steps is a “wash” of the plate using a buffer, such as phosphate-buffered saline (PBS) and a non-ionic detergent, to remove unbound material. The wells are washed two or more times during each wash step, depending on the specific protocol.[4]

In a usual ELISA protocol, a serial dilution of concentrations is placed in the wells of the plate. After the results are measured, a standard curve from the serial dilutions data is plotted with a concentration on the x-axis using a log scale and absorbance on the y-axis using a linear scale.[8]

There are four major types of ELISA:

- Direct ELISA (antigen-coated plate; screening antibody)
- Indirect ELISA (antigen-coated plate; screening antigen/antibody)
- Sandwich ELISA (antibody-coated plate; screening antigen)
- Competitive ELISA (screening antibody)

Direct ELISA

Both direct and indirect ELISAs begin with the coating of antigens to the ELISA plates.[9] The first binding step involves adding antigens to the plates, which are incubated for one hour at 37^oC or can be incubated at 4^oC overnight. Once the incubation step is completed, the next step is to wash the plates of any potential unbound antibodies and block any unbound sites on the ELISA plate using agents like BSA, ovalbumin, aprotinin, or other animal proteins.[2] This second step is crucial because it prevents the binding of any non-specific antibodies to the plate and minimizes false-positive results. After adding the buffer, the plate is rewashed, and a selected enzyme-conjugated primary detection antibody is added. The plate is further incubated for one hour.[10]

In a direct ELISA, the primary detection antibody binds directly to the protein of interest.[11] Next, the plate is rewashed to remove any unbound antibodies. A substrate/chromophore, such as alkaline phosphatase (AP) or horseradish peroxidase (HRP), is added to the plate, which results in a color change. The color change of the sample occurs by either the hydrolysis of phosphate groups from the substrate by AP or by the oxidation of substrates by HRP.[10] The advantages of using direct ELISA include eliminating secondary antibody cross-reactivity, and due to fewer steps, it is rapid compared to indirect ELISA. Its disadvantages include its low sensitivity compared to the other types of ELISA and its high cost of reaction.[3]

Indirect ELISA

The steps of the indirect ELISA are identical to the direct ELISA, except for an additional wash step and the types of antibodies added after the buffer is removed.[12] Indirect ELISA requires two antibodies, a primary detection antibody that sticks to the protein of interest and a secondary enzyme-linked antibody complementary to the primary antibody.[8] The primary antibody is added first, followed by a wash step, and then the enzyme-conjugated secondary antibody is added and incubated. After this, the steps are the same as the direct ELISA, which includes a wash step, the addition of substrate, and the detection of a color change.[13]

The indirect ELISA has a higher sensitivity when compared to the direct ELISA.[13] It is also less expensive and more flexible due to the many possible primary antibodies that can be used. The only major disadvantage of this type of ELISA is the risk of cross-reactivity between the secondary detection antibodies.[3]

Sandwich ELISA

Unlike direct and indirect ELISA, the sandwich ELISA begins with a capture antibody coated onto the wells of the plate.[8] It is termed a “sandwich” because the antigens are sandwiched between two layers of antibodies (capture and detection antibodies).[2] After adding the capture antibody to the plates, the plates are then covered and incubated overnight at 4^oC. Once the coating step is complete, the plates are washed with PBS, then buffered/blocked with BSA. The buffer washes are carried out at room temperature for at least 1 to 2 hours. Finally, the plate is washed with PBS once again before the antigen is added.[14]

The antigen of interest is added to the plates to bind to the capture antibody and incubated for 90 minutes at 37^oC. The plate is rewashed, and the primary detection antibody is added to the plate and incubated for another 1 to 2 hours at room temperature, followed by a buffer wash. Then the secondary enzyme-conjugated antibody is added and incubated for another 1 to 2 hours. The plate is rewashed, and the substrate is added to produce a color change.[2] The sandwich ELISA has the highest sensitivity among all the ELISA types.[14] The major disadvantages of this type of ELISA are the time and expense and the necessary use of “matched pair” (divalent/multivalent antigen) and secondary antibodies.[3]

Competitive ELISA

The competitive ELISA tests for the presence of an antibody specific for antigens in the test serum.[15] This type of ELISA utilizes two specific antibodies, an enzyme-conjugated antibody and another antibody present in the test serum (if the serum is positive). Combining the two antibodies into the wells will allow for competition for binding to antigens. The presence of a color change means that the test is negative because the enzyme-conjugated antibody bound the antigens (not the antibodies of the test serum). The absence of color indicates a positive test and the presence of antibodies in the test serum.[3] The competitive ELISA has a low specificity and cannot be used in dilute samples. However, the benefits are that there is less sample purification needed, it can measure a large range of antigens in a given sample, it can be used for small antigens, and it has low variability.[10]

Diagnostic Tests

Enzyme-linked immunosorbent assays are applied in many diagnostic tests.[1][3] Some of the uses of ELISA can include the following:

Detect and Measure the Presence of Antibodies in the Blood

- Autoantibodies (anti-dsDNA, anti-dsg1, ANA, etc.)
- Antibodies against infectious disease (antibacterial, antiviral, antifungal)
- Hepatitis A, B, C, HIV, etc.

Detect and Estimate the Levels of Tumor Markers

- Prostate-specific antigen (PSA)
- Carcinoembryonic Antigen (CEA)

Detect and Estimate Hormone Levels

- Luteinizing hormone
- Follicular stimulating hormone
- Prolactin
- Testosterone
- Human chorionic gonadotropin (hCG)

Tracking Disease Outbreaks

- Cholera
- HIV
- Influenza

Detecting Past Exposures

- HIV
- Lyme disease
- Hepatitis

Screening Donated Blood for Possible Viral Contaminants

- Anti-HIV-1/2
- Anti-HCV
- HBsAg

Detecting Drug Abuse

- Amphetamine
- Methamphetamine
- 3,4-methylenedioxymethamphetamine
- Cocaine
- Benzoylecgonine

Interfering Factors

Factors that can interfere with appropriate ELISA testing can occur at any phase of the testing process, beginning with specimen collection. The quality and integrity of the assay plate, coating buffer, capture antibody, blocking buffer, target antigen, detection antibody, enzyme conjugate, washes, substrate, and signal detection can all interfere with proper ELISA testing.[1] Some of the factors that can interfere with testing are the following:

- Plate Assay: the shape and quality of the wells, the material of the plate, potential pre-activation, and even or uneven coating.[2]
- Buffer: pH, contamination.[13]
- Capture and detection antibody: incubation time, temperature, specificity, titer, affinity.[10]
- Blocking buffer: cross-reactivity, concentration, contamination.[14]
- Target antigen: conformation, stability, epitopes.[3]
- Enzyme conjugate: type, concentration, function, cross-reactivity.[16]
- Washes: contamination, frequency, volume, duration, composition.[1]
- Substrate: quality/manufacturer.[2]
- Detection: instrument-dependent factors.[16]
- Reader/human error.[3]

Results, Reporting, and Critical Findings

Data gathered from ELISA tests can be quantitative, qualitative, or semiquantitative.[1] The quantitative concentration results are plotted and compared to a standard curve. The qualitative results confirm or deny the presence of a particular antigen/antibody in a sample.[2] The semiquantitative results compare the intensity of the signals, which can reach relative antigen levels in a sample.[16]

Once color changes are measured from the assay, the results are graphed either on paper or in software.[1] Typically, the graph compares optical density to log concentration, which gives a sigmoidal curve. Known concentrations give the graph's standard curve, and measurement of unknowns can then occur when sample values are compared to the linear portion of the graphed standard curve.[15]

Clinical Significance

ELISAs can be used in many settings, including rapid antibody screening tests for Human immunodeficiency virus (HIV), detection of other viruses, bacteria, fungi, autoimmune diseases, food allergens, blood typing, presence of the pregnancy hormone hCG, laboratory and clinical research, forensic toxicology, and many other diagnostic settings.[17][18]

In HIV testing, a blood or saliva specimen is collected for testing, typically using indirect ELISA-based tests.[4] The ELISA is a screening tool for HIV detection, but not diagnostic. Diagnosis requires further testing by Western blot due to potential false positives. Another virus**,** Molluscum contagiosum virus (MCV), which commonly infects the skin of children and young adults, can be detected by ELISA testing.[19] ELISA testing in this setting is currently being evaluated to assess global MCV seroprevalence.[20]

ELISA has also been used to detect desmogleins 1 and 3 and bullous pemphigoid antigen 180 autoantibodies implicated in pemphigus and bullous pemphigoid autoimmune blistering diseases, respectively.[21] In food allergy, the evolution of the ELISA has played an important role in allergy research and diagnosis. Ultrasensitive ELISA variations have been developed to detect quantities of allergens on the scale of picograms. This is important because of the life-threatening role that food allergies can have on a public health scale.[22]

Quality Control and Lab Safety

As with quantitative procedures, it is important to verify that the results of qualitative and semi-quantitative examinations are correct before reporting them to the requesting healthcare provider.[23] The laboratory must establish a quality control program for all of its qualitative and semi-quantitative tests.[24] When establishing this program, set policies, train staff, assign responsibilities, and ensure all necessary resources are available. Ensure that the recording of all quality control data is complete and that the quality manager and the laboratory director conduct an appropriate review of the information.[25]

Positive and negative controls are recommended for many qualitative and semi-quantitative tests, including some procedures that use special stains or reagents and tests with endpoints such as agglutination or color change.[26] These controls should generally be used with each test run.[27] The use of controls will also help validate a new lot number of test kits or reagents, check on temperatures of storage and testing areas, and evaluate the process when new testing personnel is carrying out the testing.[28]

Keep the following in mind when using traditional controls for qualitative or semi-quantitative tests: test control materials in the same manner as testing patient samples; use a positive and negative control, preferably once each day of testing, or at least as often as recommended by the manufacturer; choose positive controls that are close to the cut-off value of the test, to be sure the test can detect weak positive reactions; for agglutination procedures, include a weak positive control as well as a negative control and stronger positive control.[24]

Basic safety rules for laboratory conduct should be observed whenever working in a laboratory.[29] Consider all specimens, control materials, and calibrator materials as

potentially infectious. Exercise the usual precautions required for handling all laboratory reagents.[30] Disposal of all waste material should be in accordance with local guidelines. Wear gloves, a lab coat, and safety glasses when handling human blood specimens. Place all plastic tips, sample cups, and gloves that come into contact with blood in a biohazard waste container. Discard all disposable glassware into sharps waste containers.[31]

Protect all work surfaces with disposable absorbent bench top paper, discarded into biohazard waste containers weekly or whenever blood contamination occurs. Wipe all work surfaces weekly. All equipment should be regularly inspected for wear or deterioration. Equipment should be maintained according to the manufacturer's requirements, and records of certification, maintenance, or repairs should be maintained for the life of the equipment.[32] Computers and instrumentation should be labeled to indicate whether gloves should be worn. Inconsistent glove use around keyboards/keypads is a source of potential contamination. Avoid wearing jewelry in the lab, as this can pose multiple safety hazards. Safety rules for laboratory-specific operations will be provided in appropriate laboratory SOPs.[33]

Enhancing Healthcare Team Outcomes

ELISA testing is an important part of medical care and scientific research. Collaboration between scientists, laboratory technicians, phlebotomists, clinicians, nurses, and other medical professionals is necessary for appropriate specimen collection, testing, interpretation, diagnosis, and effective patient education and treatment planning. ELISA technologies continue to grow and play a major role in clinical research allowing for the development of more diagnostic and screening tests. The continued evolution of ELISA testing is promising for the future of medicine and has improved early diagnosis of HIV and pregnancy detection.

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